GSS TALK

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I want to evaluate

$$1 + 2 + \cdots = ?$$

Of course it as itself does not make sense. I claim that

$$1 + 2 + \cdots = -\frac{1}{12}.$$

Proof 1. The Riemann zeta function

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s},$$

makes sense as a face value when $\mathrm{Re}(s) > 1$. I believe one of the general exam materials in complex analysis is analytic continuation of Riemann zeta function, so we can say

$$1 + 2 + \cdots := \zeta(-1).$$

The value can be computed using the functional equation,

$$\zeta(s) = 2^s \pi^{s-1} \sin\left(\frac{\pi s}{2}\right) \Gamma(1-s)\zeta(1-s),$$

so

$$\zeta(-1) = 2^{-1}\pi^{-2} \sin\left(\frac{-\pi}{2}\right) \Gamma(2)\zeta(2) = -\frac{1}{12}.$$

□

Proof 2. Let $f = 1 + 2 + \cdots$. Then, $2f = 2 + 4 + \cdots$, so

$$-3f = f - 4f = (1 + 2 + \cdots) - 2(2 + 4 + \cdots) = 1 - 2 + 3 - 4 + \cdots.$$

So,

$$-6f = -3f + (-3f) = 1 + (1-2) - (2-3) + (3-4) - \cdots = 1 - 1 + 1 - 1 \cdots.$$

Since $1 + r + r^2 + \cdots = \frac{1}{1-r}$, $-6f = \frac{1}{1-(-1)} = \frac{1}{2}$, or $f = -\frac{1}{12}$. □

Of course, the second proof is bogus. Well, actually maybe not.

Definition 1. *Let K be a characteristic zero field. Let*

$$V_K = \{f : \mathbb{N} \to K : f \text{ eventually satisfies linear recurrence}\}.$$

Equivalently, there are $k \in \mathbb{N}$, $a_1, \cdots, a_k, b_1, \cdots, b_k \in \overline{K}^\times$, $c_1, \cdots, c_k \in \mathbb{N}_{\geq 0}$, such that

$$f(n) = \sum_{i=1}^{k} a_i b_i^n n^{c_i}, \qquad n \gg 0.$$

Let V_K^c be the subspace of compactly supported sequences, i.e. $f(n) = 0$ for $n \gg 0$.

Obviously, there is the K-linear sum operator $\Sigma : V_K^c \to K$. We would like to show rigorously that the two proofs should yield the same value by the following process.

(1) Σ can be extended to K-linear $\Sigma : V_K \to K$. It satisfies some properties which uniquely characterizes Σ.
(2) Analytic continuation yields an opeartor with the same properties.

For (1), the key is that V_K admits two kinds of rearrangement operators, "shift" and "dilatation,"

$$S : V_K \to V_K, \quad S(f)(n) = f(n+1),$$

$$D_m : V_K \to V_K, \quad D_m(f)(n) = \begin{cases} 0 & \text{if } m \nmid n \\ f\left(\frac{n}{m}\right) & \text{otherwise.} \end{cases}$$

These are rearrangement operators, so that Σ should be morally speaking invariant under these. However, we arrive at why this divergent sum is ill-behaved: if there is $\Sigma : V_K \to K$ that is S-invariant, then

$$0 = \Sigma(1,1,1,\cdots) - \Sigma(S(1,1,1,\cdots)) = \Sigma(1,1,1,\cdots) - \Sigma(0,1,1,\cdots) = \Sigma(1,0,0,\cdots) = 1.$$

Thus we can't expect Σ to be S-invariant.

But not all hope is lost. How much can we extend Σ to be S-invariant? If this is possible for some S-invariant subspace $V_K^c \subset W \subsetneq V_K$, we want to fill the arrow

$$\begin{tikzcd} V_K^c/(S-1)V_K^c \arrow[r] \arrow[d] & K \\ W/(S-1)W \arrow[ur, dashed] & \end{tikzcd}$$

and this would be no problem if the vertical arrow is an isomorphism! This would happen by snake lemma if $(S-1) : W/V_K^c \to W/V_K^c$ is an isomorphism.

Now S is an operator on W/V_K^c which is in fact <u>admissible</u>. Namely, if $f \in W/V_K^c$, then the smallest S-invariant subspace $\langle f \rangle$ containing f is finite-dimensional. This is precisely equivalent to that f eventually satisfies linear recurrence.

Definition 2. *The eigenvalues of S on $\langle f \rangle$ are called <u>exponents</u>. In fact, if*

$$f(n) = \sum_{i=1}^{k} a_i b_i^n n^{c_i}, \qquad n \gg 0,$$

then $1/b_1, \cdots, 1/b_k$ are the exponents of f.

So $(S-1)$ would be an isomorphism on W/V_K^c as long as W has no element having exponent 1. This is precisely why we can't extend Σ to contain $(1,1,\cdots)$ because it has exponent 1. Let $V_K^{\neq 1}$ be the subspace consisting of elements not having exponent 1. By above, we know that

Proposition 1. *Σ extends uniquely to an S-invariant K-linear operator $\Sigma : V_K^{\neq 1} \to K$, which is also D_2-invariant.*

Proof. Note that $V_K^{\neq 1}$ is D_2-invariant, because $D_2 f$ has exponents $\pm\sqrt{\lambda}$ for λ exponents of f. Thus, it is enough to show that ΣD_2 is S-invariant. But $\Sigma D_2 S f = \Sigma S^2 D_2 f = \Sigma D_2 f$, so we are done. $\square$

Proposition 2. *There is a unique K-linear extension $\Sigma : V_K \to K$ of the above that is D_2-invariant.*

Proof. To prove this we need to show that $D_2 - 1 : V_K/V_K^{\neq 1} \to V_K/V_K^{\neq 1}$ is isomorphism. But note that $V_K/V_K^{\neq 1}$ is identified with the space of polynomials. Note that $D_2(n^k) = \frac{1+(-1)^n}{2}\left(\frac{n}{2}\right)^k \equiv \frac{1}{2}\left(\frac{n}{2}\right)^k \pmod{V_K^{\neq 1}}$, because $(-1)^n$ makes the exponent -1. So, D_2 has eigenvalues $\frac{1}{2}, \frac{1}{4}, \cdots$, which makes $D_2 - 1$ invertible. $\square$

Thus, we have proved that there is a unique K-linear extension $\Sigma : V_K \to K$ of $V_K^c \to K$ which is D_2-invariant and $\Sigma|_{V_K^{\neq 1}}$ is S-invariant. Such characterization is unique for any $W \subset V_K$ which is S-invariant and D_2-invariant.

Now we consider analytic continuation. For $f \in V_{\mathbb{C}}$, we want to consider $\Sigma(f)$ by analytic continuation. This involves $\sum_n \frac{f(n)}{n^s}$. Note that first of all the function has no region of absolute convergence if there is an exponent whose absolute value is < 1 (so that $f(n)$ is exponentially growing). So we can only do this for $f \in V_{\mathbb{C}}^{\geq 1}$, when the absolute values of exponents are all ≥ 1.

Proposition 3. *For $f \in V_{\mathbb{C}}^{\geq 1}$, $\zeta_f(s) := \sum_n \frac{f(n)}{n^s}$ has meromorphic continuation to $\mathbb{C}$, and $\zeta_f(0)$ is well-defined and is equal to $\Sigma(f)$.*

Proof. We prove the first claim. First, f is a sum of polynomial and an element of $V_{\mathbb{C}}^{>1}$. For a polynomial we know how to continue: zeta functions. So we assume $f \in V_{\mathbb{C}}^{>1}$. As $f \in V_{\mathbb{C}}^{>1}$, $f(n)$ is bounded by a degree N polynomial. So, $\zeta_f(s)$ is absolutely convergent for $\operatorname{Re} s > 1 + N$. We use the trick that $S - 1$ is invertible on $\langle f \rangle \subset V_{\mathbb{C}}^{>1}/V_{\mathbb{C}}^c$. Thus, $f = (S-1)f' + f''$ for $f' \in \langle f \rangle + V_{\mathbb{C}}^c$ and $f'' \in V_{\mathbb{C}}^c$. Then

$$\sum_n \frac{f(n)}{n^s} = \sum_n f'(n)\left(\frac{1}{(n+1)^s} - \frac{1}{n^s}\right) + \sum_n \frac{f''(n)}{n^s}.$$

The second part is analytic, while the first part has absolute convergence region $\operatorname{Re} s > N$, because $\frac{1}{(n+1)^s} - \frac{1}{n^s} \sim \frac{1}{n^{s+1}}$ and $f' \in \langle f \rangle$ means f' is also bounded by a degree N polynomial. One can do this iteratedly to obtain a meromorphic continuation.

Now we prove the second claim. Since f is bounded by a polynomial, we know $\zeta_f(0)$ is bounded by a linear combination of $\zeta(0), \zeta(-1), \cdots$ which are all some values, so $\zeta_f(s)$ has no pole at $s = 0$. Finally, to prove the third claim, we need to prove that $f \mapsto \zeta_f(0)$ is D_2-invariant and S-invariant on $V_{\mathbb{C}}^{>1}$. D_2-invariance is easy, as

$$\zeta_{D_2 f}(s) = \sum_n \frac{D_2 f(n)}{n^s} = \sum_n \frac{f(n)}{(2n)^s} = 2^{-s}\zeta_f(s).$$

To prove S-invariance for $f \in V_{\mathbb{C}}^{>1}$, we compute $\zeta_{Sf}(s) - \zeta_f(s)$. Note that

$$\begin{aligned}
\zeta_{Sf}(s) - \zeta_f(s) &= \sum_n f(n)\left(\frac{1}{(n+1)^s} - \frac{1}{n^s}\right) \\
&= \sum_n \frac{f(n)}{n^s}((1+1/n)^{-s} - 1) \\
&= \sum_n \frac{f(n)}{n^s} \sum_{k \geq 1} \binom{-s}{k} \frac{1}{n^k}
\end{aligned}$$

Now say $f(n)$ is bounded by a polynomial of degree N. Then $\sum_n \frac{\binom{-s}{k}}{s} \frac{f(n)}{n^{s+k}}$ is absolutely convergent for $\operatorname{Re} s > 1 + N - k$. Now we can cut the tail of Taylor series for $(1 + 1/n)^{-s}$ so that the

remainder term has small error due to Taylor inequality. Thus it would follow that $\frac{\zeta_{Sf}(s)-\zeta_f(s)}{s}$ has analytic continuation to a small neighborhood of real segment $[0, 1+N]$ without any poles. Thus $\zeta_{Sf}(0) = \zeta_f(0)$. $\square$

Now we can prove the following.

Proposition 4. *For a Dirichlet character χ, $L(-k,\chi) \in \mathbb{Q}(\chi)$ for k nonnegative integer.*

Proof. Since $\chi(n)n^k \in V_{\mathbb{C}}^{\geq 1}$, $L(-k,\chi) = S(\chi(n)n^k)$. But S is $\mathbb{Q}(\chi)$-valued. $\square$

The point is that all remainder terms are divisible by s, so you get 0 at $s=0$. (That is what breaks down if there are poles.) To make this rigorous, we can write

$$\frac{(1+1/n)^{-s}-1}{s} = -\frac{1}{n} + \ldots + \frac{P_{k-1}(s)}{n^{k-1}} + R_k(s,n).$$

As f is bounded by a polynomial, we may choose k large enough so that $|f(n)| \leq Cn^{k-2}$. Then $\sum \frac{f(n)}{n^s}$ is absolutely convergent for $\operatorname{Re} s \geq k$. Let's go back to the expression

$$I(s) = s\sum_n \frac{f(n)}{n^s}\left(-\frac{1}{n} + \ldots + \frac{P_{k-1}(s)}{n^{k-1}} + R_k(s,n)\right).$$

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For $|s| \leq k+1$, we have a Taylor estimate $|R_k(s,n)| \leq \frac{C_k}{n^k}$. So the series is absolutely convergent near $s=k$. Hence, at least in an open neighborhood of $s=k$, we have

$$I(s) = s\left(-\sum \frac{f(n)}{n^{s+1}} + \ldots + P_{k-1}(s)\sum \frac{f(n)}{n^{s+k-1}} + \sum f(n)R_k(s,n)\right)$$

and the term $\sum f(n)R_k(s,n)$ is analytic in $|s| \leq k+1, \operatorname{Re} s > -1/2$. The other terms all admit analytic continuations, by our preceding discussion. So finally we can *rigorously* say that the right hand side is "divisible by s," hence $I(0) = 0$. $\square$